

Bail(ey)ing Out

Bailey Hall is one of the largest public spaces at Cornell University. Used for academic events, concerts, and convocations, this stately building is shaped as an amphitheatre: a semicircular auditorium, with a corridor and colonnaded portico wrapping around it, and a main entrance through a foyer on the south side. As with any building on campus, the number of people allowed in the Bailey Hall auditorium is limited not only by acoustics and convenience, but also by the NY State Fire Code – to ensure a quick and orderly exit, particularly in case of emergency. While the University already has an excellent evacuation and safety plan in place, ***your team's goal is to explore whether leaving this building can be made both easier and faster*** without major changes to its infrastructure and appearance. To do this, you will ideally consider both emergency evacuations and ordinary departures (e.g., the audience flowing out after a large lecture or concert).

Crowd motion can be complicated even without any emergency. People make their individual choices (*which exit to go to? along which path? how quickly? what to do if there is a traffic jam or the preferred route is somehow blocked?*). They make those choices individually, but clearly affect each other in the process. Some of them might be brutally rational (*what's the quickest way out?*), others may be influenced by habit (*how did I enter this room?*), peer pressure (*which way is everyone going?*), or social norms (*it would be rude to rush ahead of others!*). The majority operate with only partial knowledge of their environment (*which doors are really open right now? how many people will choose each of them?*). What appears to be the best strategy for each individual might end up impeding the motion of the whole group. Traffic engineers know that, when it comes to independent decision makers, restricting the range of choices they have can sometimes improve the outcomes for everyone. This is particularly important in evacuation scenarios, where Bailey Hall would have to be emptied in under 180

seconds – from the moment when the emergency is declared and until the last person leaves the building. For large non-academic events (e.g., concerts or performances involving more than 500 people), the state law requires the presence of at least one "Crowd Manager" for every 250 attendees. Any such event that fills all 1,329 seats in Bailey Hall is thus required to include at least six Crowd Managers – specially trained ushers, who can provide detailed evacuation instructions to the audience. However, guiding and assisting that many people is difficult, particularly if this is not a drill but a real emergency. Can we make their jobs easier through minor changes to the environment?

Develop and use a mathematical model to address (some of) the following questions:

1) How long does it take for the audience to leave Bailey Hall after a concert?

2) How long would it take for the same concert audience to leave in case of emergency while guided by Crowd Managers? How would that egress time change in case of an unexpected (and possibly initially unknown) obstruction of one of the exits?

3) How would the above differ from the time needed to evacuate the same number of students from a large lecture taught in Bailey Hall? Note that Crowd Managers are not present during academic events since students are very familiar with their regular lecture buildings. They also participate in fire drills and are assumed to know the location of emergency exits.

4) Could these building egress times be decreased safely
    a) by using temporary/removable barriers in the corridor to restrict people's paths from the mezzanine exits to specific exterior doorways?
    b) or by attaching directional LED light strips to the floors or walls to guide different parts of the audience to specific exits?

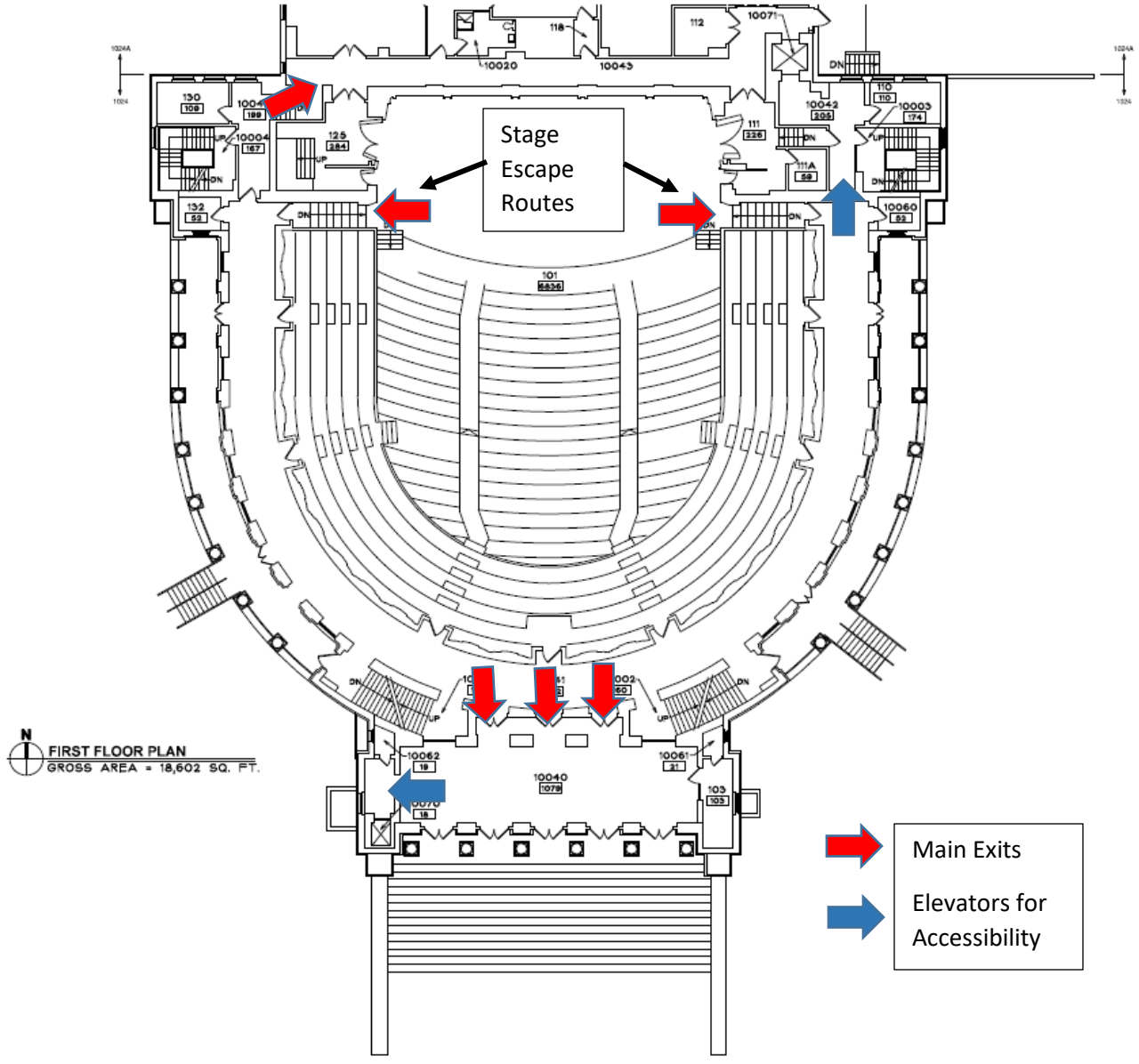
    (For concerts, both of these measures are meant to help Crowd Managers rather than replace them. Their minimum number is mandated by the state law.)

5) Would such measures be cost-effective? E.g., how would their installation and maintenance costs compare with the cost of having more Crowd Managers on duty?

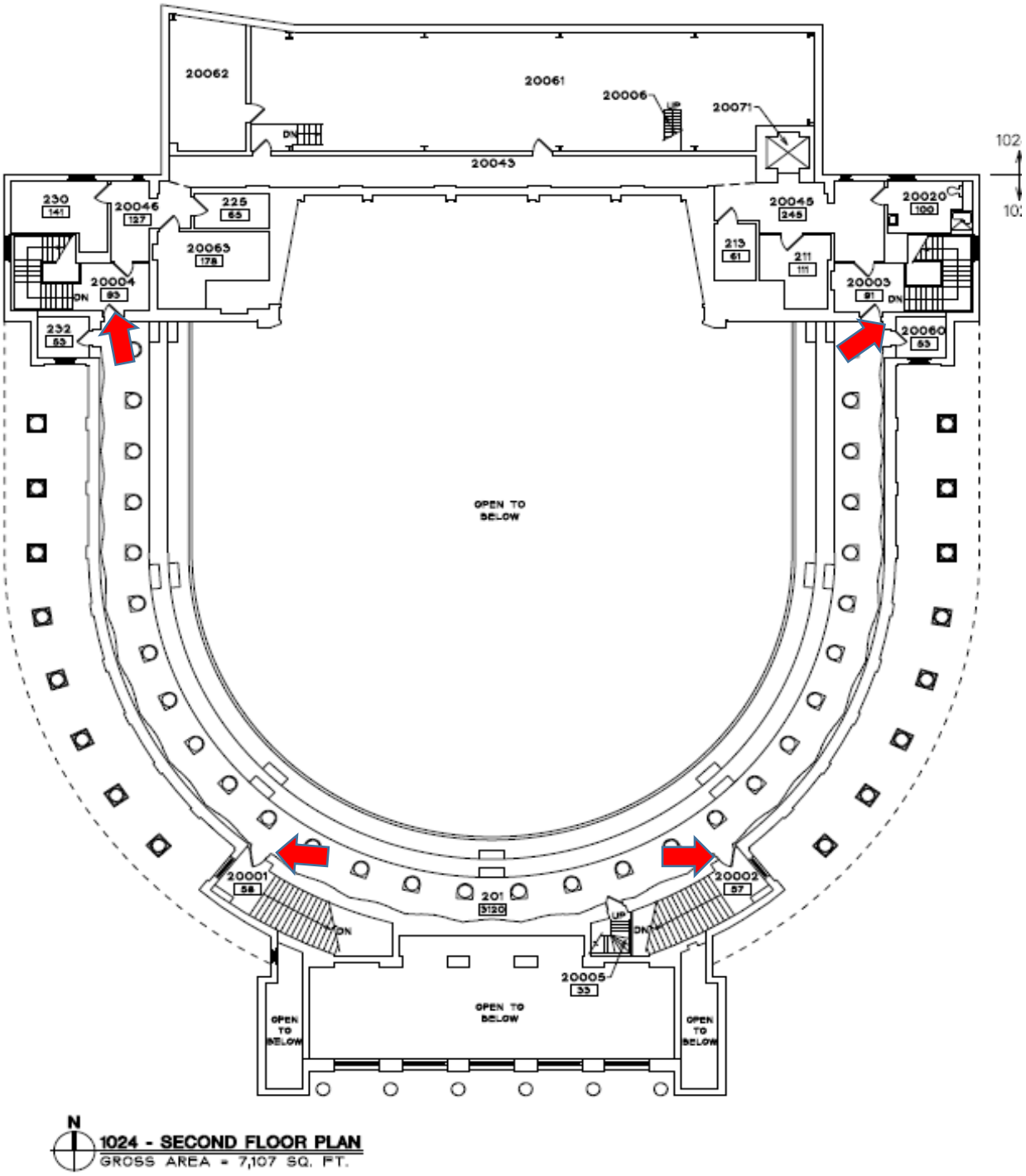
While working on these questions, you should keep in mind that good evacuation guidance should be robust to execution error. A good plan must balance efficiency and simplicity since an erroneous implementation by confused/scared people might easily increase the egress times.

In addition to your technical description of the model and the results, write a one-page executive summary of your findings addressed to Mr. Timothy Ryan, University Fire Marshal and Director of Fire Protection and Emergency Services.

Main Auditorium – First Floor



Main Auditorium – Balcony

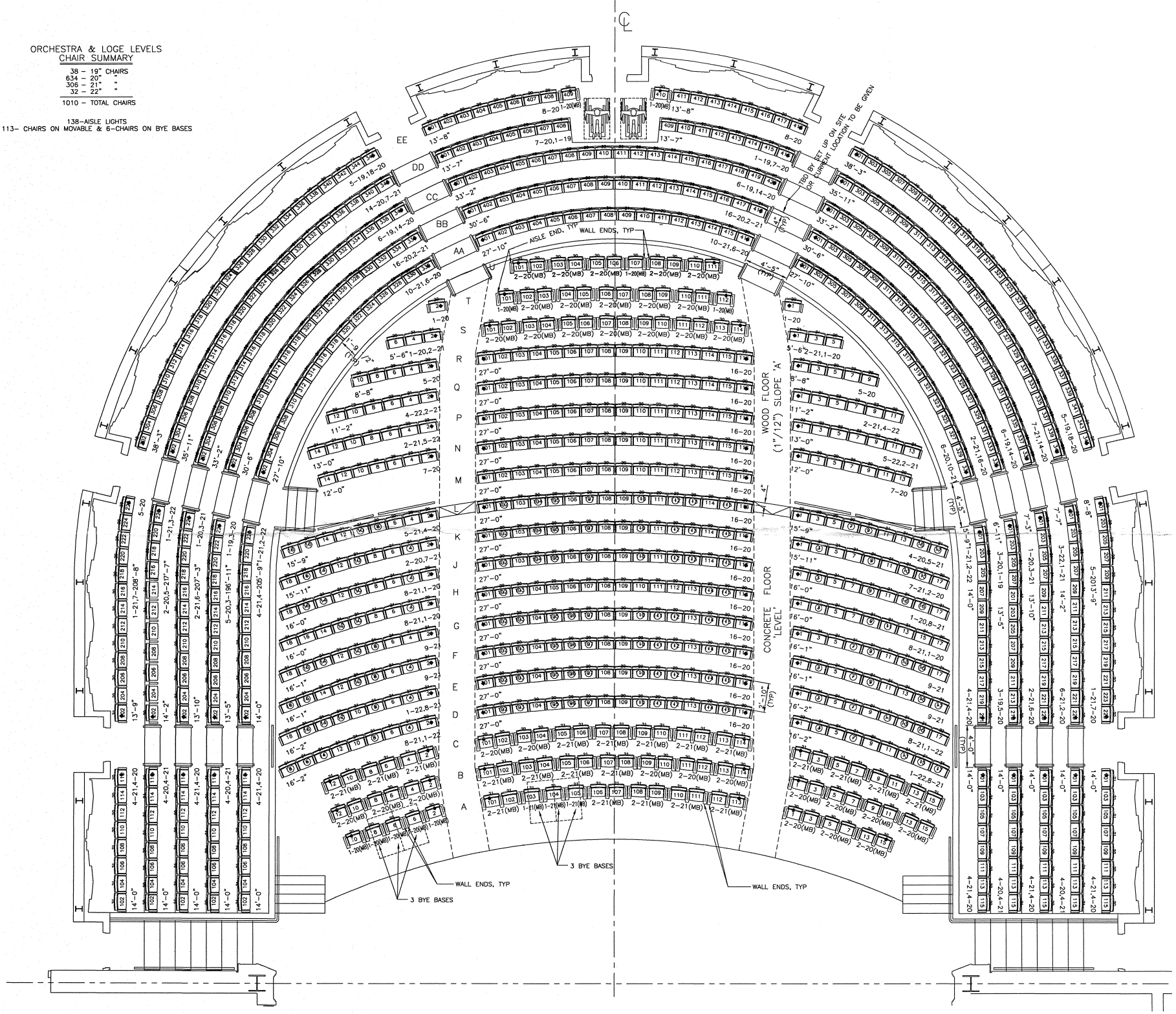


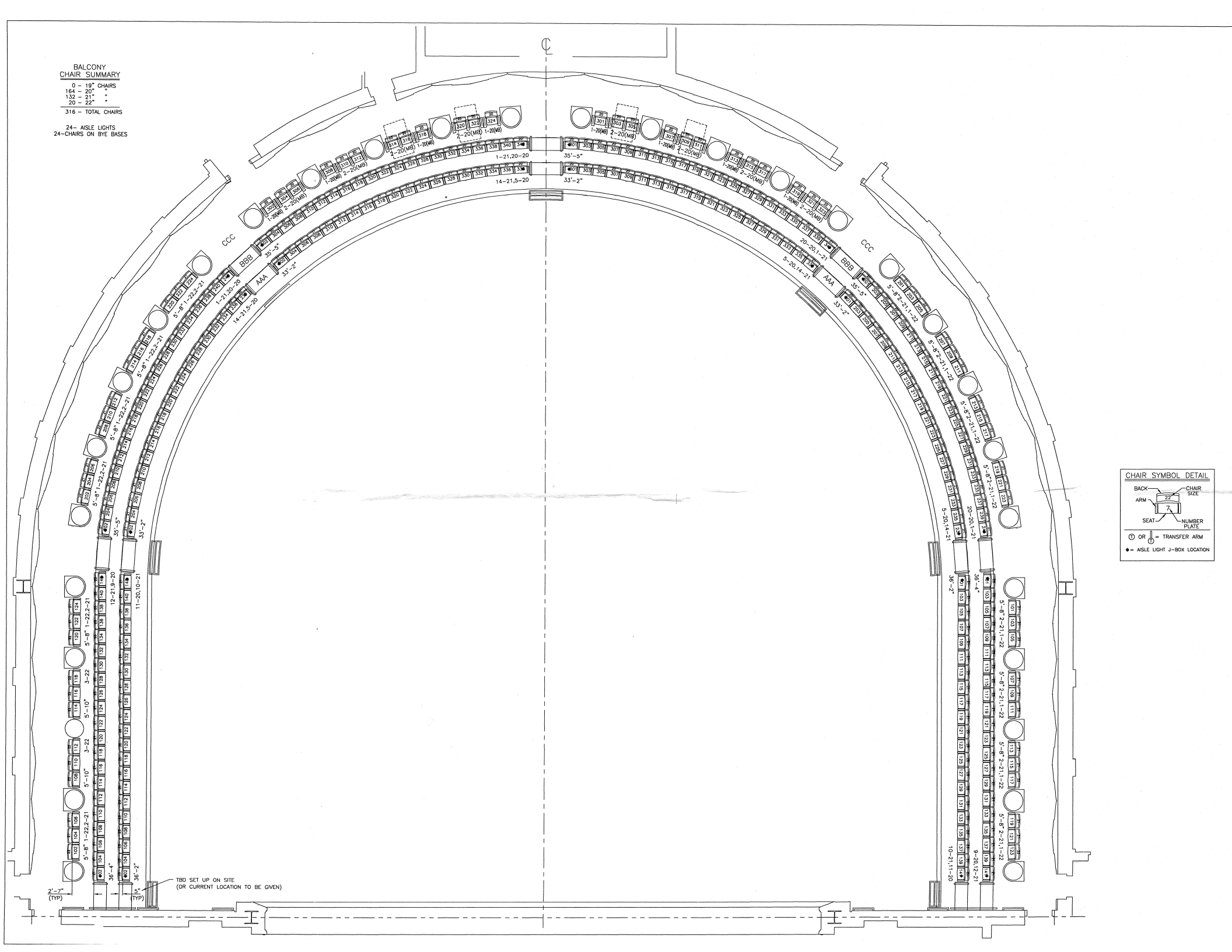
ORCHESTRA & LOGE LEVELS
CHAIR SUMMARY

38 – 19" CHAIRS
634 – 20" "
306 – 21" "
32 – 22" "

1010 – TOTAL CHAIRS

138–AISLE LIGHTS
113– CHAIRS ON MOVABLE & 6–CHAIRS ON BYE BASES




BALCONY
CHAIR SUMMARY
0 – 19" CHAIRS
164 – 20" "
132 – 21" "
20 – 22" "
316 – TOTAL CHAIRS
24– AISLE LIGHTS
24–CHAIRS ON BYE BASES
CCC
BBB
AAA
1–21,20–20
14–21,5–20
1–20(MB)
2–20(MB)
35'–5"
33'–2"
5'–8" 1–22,2–21
5'–8" 2–21,1–22
20–20,1–21
5–20,14–21
12–21,9–20
11–20,10–21
10–21,11–20
9–20,12–21
3–22
5'–10"
36'–4"
36'–2"
2'–7" (TYP)
5" (TYP)
TBD SET UP ON SITE
(OR CURRENT LOCATION TO BE GIVEN)
CHAIR SYMBOL DETAIL
BACK
CHAIR SIZE
ARM
SEAT
NUMBER PLATE
OR = TRANSFER ARM
= AISLE LIGHT J–BOX LOCATION